WEEK 10: LINEAR PROGRAMMING II — DUALITY

Advanced Algorithms

ROADMAP

1. Certifying an optimum: bounding an LP by combining its constraints
2. The dual LP, and a recipe for writing it
3. Weak duality (proved) and strong duality (stated)
4. Max-flow = min-cut, round two: as LP duality
5. The matching capstone: Kőnig's theorem
6. Complementary slackness — and what a negative price tells you

PART I — CERTIFICATES AND BOUNDS

HOW DO WE *CERTIFY* AN LP OPTIMUM?

Last week we saw an optimum of a (feasible, bounded) LP occurs at a vertex. A skeptic asks: *how do I know no feasible point does better?*

We have played this game before:

- **Week 8:** to certify a flow is maximum, exhibit a *cut* of the same value — no need to re-run Ford–Fulkerson.

HOW DO WE *CERTIFY* AN LP OPTIMUM?

Last week we saw an optimum of a (feasible, bounded) LP occurs at a vertex. A skeptic asks: *how do I know no feasible point does better?*

We have played this game before:

- **Week 4:** greedy picks k disjoint activities. For half-open intervals $[s, f)$, choose one point sufficiently close before each selected finish time. These k points *hit every activity* — so no $k + 1$ disjoint activities exist.

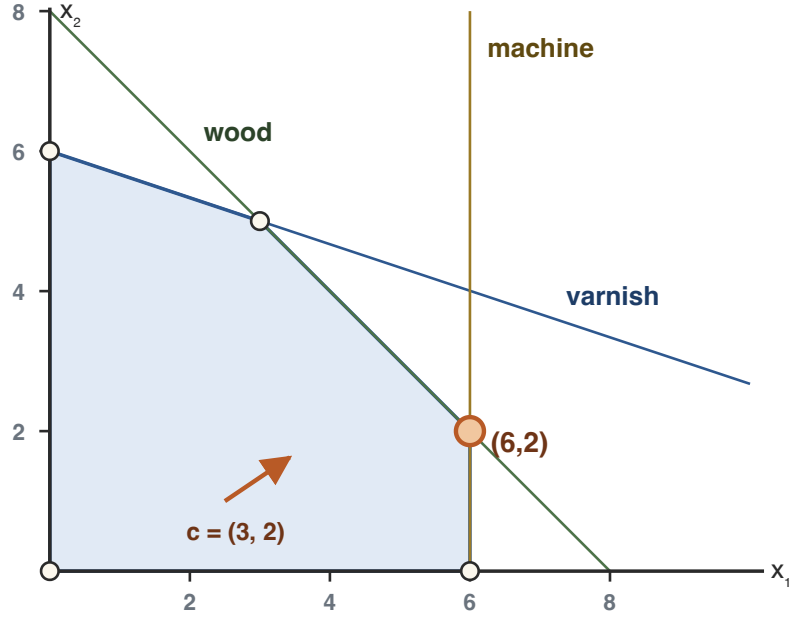
HOW DO WE *CERTIFY* AN LP OPTIMUM?

GOAL A short, checkable *certificate of optimality* for any LP.
Today's answer: a solution to a second LP — the **dual**.

LAST WEEK'S LP

$$
\begin{aligned}
\max \quad & 3x_1 + 2x_2 \\
\text{s.t.} \quad & x_1 \leq 6 \quad \text{(machine)} \\
& x_1 + x_2 \leq 8 \quad \text{(wood)} \\
& x_1 + 3x_2 \leq 18 \quad \text{(varnish)} \\
& x_1, x_2 \geq 0
\end{aligned}
$$

Chairs x_1, tables x_2.



The feasible region and the claimed optimum $(6, 2)$.

LAST WEEK'S LP

Same LP as Week 9, with (x, y) renamed (x_1, x_2) — the letter y is about to get a new job.

The sweep found $(6, 2)$, value $3 \cdot 6 + 2 \cdot 2 = 22$.

But *checking all vertices* is not a certificate that scales. We want an upper bound proof.

BOUNDING FROM ABOVE

$$\max\ 3x_1 + 2x_2 \quad \text{s.t.}$$

$$\underbrace{x_1 \le 6}_{\text{machine}},\quad \underbrace{x_1 + x_2 \le 8}_{\text{wood}},\quad \underbrace{x_1 + 3x_2 \le 18}_{\text{varnish}},\quad x_1, x_2 \ge 0$$

Idea: **combine the constraints**. Multiply each by a number $y_i \ge 0$ and add. If the result dominates the objective coefficient-wise, its right-hand side is an upper bound.

Attempt 1 — take $2\times$ (machine) $+\ 1\times$ (varnish):

$$2x_1 + (x_1 + 3x_2) \;=\; 3x_1 + 3x_2 \;\le\; 12 + 18 \;=\; 30.$$

Since $x_2 \ge 0$: $3x_1 + 2x_2 \le 3x_1 + 3x_2 \le 30$. Bound: 30.

BOUNDING FROM ABOVE

$$\max\ 3x_1 + 2x_2 \quad \text{s.t.}$$

$$\underbrace{x_1 \le 6}_{\text{machine}},\quad \underbrace{x_1 + x_2 \le 8}_{\text{wood}},\quad \underbrace{x_1 + 3x_2 \le 18}_{\text{varnish}},\quad x_1, x_2 \ge 0$$

Attempt 2 — take $3\times$ (wood):

$$3x_1 + 3x_2 \le 24.$$

Same left side as Attempt 1, cheaper right side. Bound: 24.

Getting closer to 22. Both attempts overshoot the x_2 coefficient ($3 > 2$). Can a mix avoid that waste?

YOUR TURN: A BOUND OF 22

$$
\begin{aligned}
x_1 &\le 6 && \text{(machine)} \\
x_1 + x_2 &\le 8 && \text{(wood)} \\
x_1 + 3x_2 &\le 18 && \text{(varnish)}
\end{aligned}
$$

CHALLENGE Combine these constraints with nonnegative weights to prove $3x_1 + 2x_2 \le 22$ for every feasible point.

Hint: try using just machine and wood. What weights give exactly $3x_1 + 2x_2$?

THE PERFECT COMBINATION

Attempt 3 — take $1\times$ (machine) $+\ 2\times$ (wood):

$$x_1 + 2(x_1 + x_2) \;=\; 3x_1 + 2x_2 \;\le\; 6 + 16 \;=\; 22.$$

This time the combination is *exactly* the objective — no slack wasted. Bound: 22.

And the feasible point $(6, 2)$ *achieves* 22.

> **CERTIFIED** Every feasible point has value ≤ 22, and $(6, 2)$ has value 22. The pair — point $(6, 2)$, multipliers $(y_1, y_2, y_3) = (1, 2, 0)$ — is a two-line proof of optimality anyone can check.

THE BEST BOUND IS AN LP

$$
\begin{aligned}
y_1 \times \quad & x_1 \le 6 && (\text{machine}) \\
y_2 \times \quad & x_1 + x_2 \le 8 && (\text{wood}) \\
y_3 \times \quad & x_1 + 3x_2 \le 18 && (\text{varnish}) \\
\text{objective:} \quad & 3x_1 + 2x_2
\end{aligned}
$$

- **Legal:** $y_1, y_2, y_3 \ge 0$ (adding $\le$-inequalities only works with nonnegative weights).
- **Dominates the objective:** coefficient of x_1: $y_1 + y_2 + y_3 \ge 3$; coefficient of x_2: $y_2 + 3y_3 \ge 2$.
  (Domination suffices because $x_1, x_2 \ge 0$.)
- **Bound obtained:** $6y_1 + 8y_2 + 18y_3$ — make it as small as possible.

THE BEST BOUND IS AN LP

Primal

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 \\ & x_1 \leq 6 \\ & x_1 + x_2 \leq 8 \\ & x_1 + 3x_2 \leq 18 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Dual

$$\begin{aligned} \min \quad & 6y_1 + 8y_2 + 18y_3 \\ & y_1 + y_2 + y_3 \geq 3 \\ & y_2 + 3y_3 \geq 2 \\ & y_1, y_2, y_3 \geq 0 \end{aligned}$$

Three constraints $\rightarrow$ three dual variables; two primal variables $\rightarrow$ two dual constraints.

THE BEST BOUND IS AN LP

Definition The search for the best upper bound is itself a linear program: the **dual** of the original (**primal**) LP.

PART II — THE DUAL RECIPE

RUNNING EXAMPLE, IN MATRICES

$$\max\ 3x_1 + 2x_2 \quad \text{s.t.}$$
$$\underbrace{x_1 \le 6}_{\text{machine}},\quad \underbrace{x_1 + x_2 \le 8}_{\text{wood}},\quad \underbrace{x_1 + 3x_2 \le 18}_{\text{varnish}},\quad x_1, x_2 \ge 0$$

The same LP in matrix form:

$$\max\ \underbrace{\begin{pmatrix}3 & 2\end{pmatrix}}_{c^\mathsf{T}} \underbrace{\begin{pmatrix}x_1\\ x_2\end{pmatrix}}_{x} \quad \text{s.t.} \quad \underbrace{\begin{pmatrix}1 & 0\\ 1 & 1\\ 1 & 3\end{pmatrix}}_{A} \begin{pmatrix}x_1\\ x_2\end{pmatrix} \le \underbrace{\begin{pmatrix}6\\ 8\\ 18\end{pmatrix}}_{b},\quad x \ge 0$$

COMBINING CONSTRAINTS

Multiply the constraints by $y_1, y_2, y_3 \geq 0$ and add. The left side is

$$\begin{pmatrix} y_1 & y_2 & y_3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (y_1 + y_2 + y_3)\, x_1 + (y_2 + 3y_3)\, x_2.$$

COMBINING CONSTRAINTS

We want the combination to dominate $3x_1 + 2x_2$, coefficient by coefficient:

$$\begin{pmatrix} y_1 & y_2 & y_3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \end{pmatrix} \geq \begin{pmatrix} 3 & 2 \end{pmatrix}, \quad \text{i.e.} \quad y^{\mathsf{T}} A \geq c^{\mathsf{T}}.$$

Multiplied out: $y_1 + y_2 + y_3 \geq 3$ and $y_2 + 3y_3 \geq 2$ — the dual constraints from Part I.

THE BOUND, IN ONE LINE

Claim If $y \geq 0$ and $y^{\mathsf{T}} A \geq c^{\mathsf{T}}$, then every feasible x satisfies

$$c^{\mathsf{T}} x \;\leq\; y^{\mathsf{T}} A x \;\leq\; y^{\mathsf{T}} b.$$

- First $\leq$: $y^{\mathsf{T}} A$ dominates c^{T} entrywise, and $x \geq 0$.
- Second $\leq$: $Ax \leq b$, weighted by $y \geq 0$.
- Attempt 3 was $y = (1, 2, 0)$: $y^{\mathsf{T}} A = (3, 2) = c^{\mathsf{T}}$, $y^{\mathsf{T}} b = 6 + 16 = 22$.

DUAL OF THE STANDARD FORM

Nothing used the specific numbers. Any LP in Week 9's standard form ($A \in \mathbb{R}^{m \times n}$)

$$\max\ c^{\mathsf{T}} x \quad \text{s.t.} \quad Ax \leq b,\ \ x \geq 0$$

has the same bound: $c^{\mathsf{T}} x \leq y^{\mathsf{T}} b$ for every $y \in \mathbb{R}^m$ with $y \geq 0$ and $y^{\mathsf{T}} A \geq c^{\mathsf{T}}$.

Transposing, $y^{\mathsf{T}} A \geq c^{\mathsf{T}}$ is the same as $A^{\mathsf{T}} y \geq c$: one constraint per column of A.

DUAL OF THE STANDARD FORM

Best bound = smallest right-hand side:

DUAL LP

$$\min\ b^\mathsf{T} y \quad \text{s.t.} \quad A^\mathsf{T} y \geq c,\ \ y \geq 0.$$

Constraints of the primal $\leftrightarrow$ variables of the dual, and vice versa; A becomes A^T; b and c trade places.

MIXED CONSTRAINTS

Real LPs mix $\leq$ rows and $=$ rows:

$$\begin{aligned} \max \quad & 3x_1 + x_2 \\ \text{s.t.} \quad & x_1 + x_2 = 4 \\ & x_1 - x_2 \leq 2 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Combining constraints still works, but the signs need care: may the equality get a *negative* multiplier?

One idea answers this: **penalise, don't enforce**.

PENALISE, DON'T ENFORCE

Keep the sign constraints $x_1, x_2 \geq 0$. Drop the other constraints and add a term per row to the objective, with a price y_i:

$$L(x, y) = 3x_1 + x_2 + y_1 (4 - x_1 - x_2) + y_2 (2 - x_1 + x_2).$$

GOAL L never undercuts the objective on feasible points:

$$3x_1 + x_2 \;\leq\; L(x, y) \quad \text{for every feasible } x.$$

Then $\max_x L(x, y)$ bounds OPT from above. Choose the signs of y to make this hold.

PENALISE, DON'T ENFORCE

$$L(x, y) = 3x_1 + x_2 + y_1\,(4 - x_1 - x_2) + y_2\,(2 - x_1 + x_2)$$

Goal: every added term is ≥ 0 on feasible points.

- Feasible x has $4 - x_1 - x_2 = 0$: the term vanishes, so y_1 may have *any sign*.
- Feasible x has $2 - x_1 + x_2 \geq 0$: we need $y_2 \geq 0$.
- Now a violated $\leq$ row ($x_1 - x_2 > 2$) makes its term negative. We are maximising, so that is a **penalty**: y_2 per unit of violation.

EVERY y GIVES A BOUND

For y_1 free and $y_2 \geq 0$:

$$\mathrm{OPT} \;\leq\; \max_{x_1, x_2 \geq 0} L(x, y).$$

The max on the right is over *all* such x, feasible or not — that only makes it larger.

Regroup L by variable:

$$\begin{aligned} L(x, y) &= 3x_1 + x_2 + y_1 (4 - x_1 - x_2) + y_2 (2 - x_1 + x_2) \\ &= 4y_1 + 2y_2 + (3 - y_1 - y_2)\, x_1 + (1 - y_1 + y_2)\, x_2. \end{aligned}$$

WHEN IS THE BOUND FINITE?

$$L(x, y) = 4y_1 + 2y_2 + (3 - y_1 - y_2)\, x_1 + (1 - y_1 + y_2)\, x_2$$

A bound of $+\infty$ is useless. Which y keep $\max_x L(x, y)$ finite?

- A positive coefficient on x_1 lets $x_1 \to \infty$. Need $y_1 + y_2 \geq 3$.
- Likewise for x_2. Need $y_1 - y_2 \geq 1$.
- With both, each x_j term is ≤ 0 for every $x_j \geq 0$ — largest at $x = 0$. So

$$\max_{x_1, x_2 \geq 0} L(x, y) = 4y_1 + 2y_2.$$

THE BEST BOUND

Fix a y with $y_2 \geq 0$ that satisfies the constraints from the previous slide. Then for every feasible x:

$$3x_1 + x_2 \;\leq\; L(x, y) \;\leq\; \max_{x'} L(x', y) \;=\; 4y_1 + 2y_2.$$

Every such y bounds OPT. The best bound is the *smallest*:

DUAL — DERIVED, NOT LOOKED UP

$$\min \; 4y_1 + 2y_2 \quad \text{s.t.}$$
$$y_1 + y_2 \geq 3, \;\; y_1 - y_2 \geq 1, \;\; y_1 \text{ free}, \; y_2 \geq 0$$

DO THE BOUNDS MEET?

Primal

$$\begin{aligned} \max \quad & 3x_1 + x_2 \\ & x_1 + x_2 = 4 \\ & x_1 - x_2 \le 2 \\ & x_1, x_2 \ge 0 \end{aligned}$$

Dual

$$\begin{aligned} \min \quad & 4y_1 + 2y_2 \\ & y_1 + y_2 \ge 3 \\ & y_1 - y_2 \ge 1 \\ & y_1 \text{ free}, \; y_2 \ge 0 \end{aligned}$$

Check with numbers. Primal: $x_2 = 4 - x_1$ and $x_1 - x_2 \le 2$ give $x_1 \le 3$; optimum $x^\star = (3, 1)$, value $\mathbf{10}$.

Dual: $y^\star = (2, 1)$ is feasible ($3 \ge 3, 1 \ge 1$), value $8 + 2 = \mathbf{10}$. They meet!

THE LAGRANGIAN

General primal: $\max\ c^{\mathsf{T}}x$ subject to rows $a_i^{\mathsf{T}}x \le b_i$ or $a_i^{\mathsf{T}}x = b_i$, and $x \ge 0$. The **Lagrangian** adds price y_i times the slack of row i:

$$L(x, y) = c^{\mathsf{T}}x + \sum_i y_i\,(b_i - a_i^{\mathsf{T}}x) = c^{\mathsf{T}}x + y^{\mathsf{T}}(b - Ax).$$

Goal: $c^{\mathsf{T}}x \le L(x, y)$ for every feasible x, i.e. every term $y_i\,(b_i - a_i^{\mathsf{T}}x) \ge 0$:

- $a_i^{\mathsf{T}}x \le b_i$: the slack is ≥ 0, so $y_i \ge 0$.
- $a_i^{\mathsf{T}}x = b_i$: the slack is 0, so y_i is unrestricted.

THE LAGRANGIAN

Regroup by variable:

$$L(x, y) = b^{\mathsf{T}} y + (c - A^{\mathsf{T}} y)^{\mathsf{T}} x = b^{\mathsf{T}} y + \sum_j \left(c_j - (A^{\mathsf{T}} y)_j\right) x_j.$$

The max over x is finite exactly when no x_j can push L to $+\infty$:

- Since $x_j \geq 0$, its coefficient must be ≤ 0: $\quad (A^{\mathsf{T}} y)_j \geq c_j$, i.e. $A^{\mathsf{T}} y \geq c$.
- Then every x_j term is ≤ 0, the max is at $x = 0$, and it equals $b^{\mathsf{T}} y$.
- Best bound: **minimise** $b^{\mathsf{T}} y$ over these y — the dual.

Sanity check: standard form ($Ax \leq b, x \geq 0$) gives $\min\ b^{\mathsf{T}} y$ s.t. $A^{\mathsf{T}} y \geq c$, $y \geq 0$ — the dual from before.

DUAL OF THE DUAL

FACT For standard form, the dual of the dual is the primal: negate the dual to get a standard-form max, take its dual, negate back.

$$\begin{aligned}
&\min\{b^\mathsf{T} y : A^\mathsf{T} y \geq c,\ y \geq 0\} \\
&= -\max\{-b^\mathsf{T} y : -A^\mathsf{T} y \leq -c,\ y \geq 0\} \\
&\xrightarrow{\text{dual}} \quad -\min\{-c^\mathsf{T} x : -Ax \geq -b,\ x \geq 0\} \\
&= \max\{c^\mathsf{T} x : Ax \leq b,\ x \geq 0\}.
\end{aligned}$$

DUAL OF THE DUAL

Consequences:

- Duality is a genuine *pairing*, not a one-way construction — each LP certifies the other.
- "Primal" and "dual" are roles, not properties: solve or analyse whichever of the pair is more convenient.

PART III — WEAK AND STRONG DUALITY

WEAK DUALITY

Theorem (weak duality) If x is feasible for the primal $\max\{c^\mathsf{T}x : Ax \le b,\ x \ge 0\}$ and y is feasible for its dual, then $c^\mathsf{T}x \le b^\mathsf{T}y$.

Proof (one line):

$$c^\mathsf{T}x \ \le\ (A^\mathsf{T}y)^\mathsf{T}x \ =\ y^\mathsf{T}(Ax) \ \le\ y^\mathsf{T}b. \qquad \blacksquare$$

Note: the proof is exactly the "combine the constraints" trick from Part I.

WEAK DUALITY

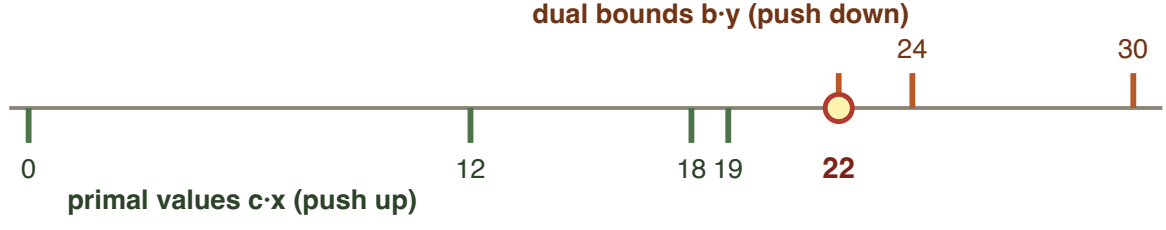
Unpacking each step of $c^\mathsf{T} x \leq (A^\mathsf{T} y)^\mathsf{T} x = y^\mathsf{T}(Ax) \leq y^\mathsf{T} b$:

- **First $\leq$:** $A^\mathsf{T} y \geq c$ coefficient-wise (dual feasibility), and $x \geq 0$ — multiplying a bigger coefficient by a nonnegative number can only increase the sum.
- **Middle $=$:** just re-bracketing, $(A^\mathsf{T} y)^\mathsf{T} x = y^\mathsf{T} A x = y^\mathsf{T}(Ax)$.
- **Second $\leq$:** $Ax \leq b$ row-wise (primal feasibility), and $y \geq 0$.

CERTIFICATES FOR FREE

COROLLARY If $x^\star$ is primal feasible, $y^\star$ is dual feasible, and $c^\mathsf{T} x^\star = b^\mathsf{T} y^\star$, then **both are optimal**.

Every primal value sits below every dual value — if they touch, neither side can move.



The running example: vertex values $0, 12, 18, 19$ from below; Attempts 1–3 give bounds $30, 24, 22$ from above; they meet at 22.

This is the same logic as Week 8's lecture: a flow of value v plus a cut of capacity v certify each other.

STRONG DUALITY

Theorem (strong duality — stated, not proved) If the primal has a finite optimum, so does the dual, and the two optimal values are **equal**:

$$\max\{c^{\mathsf{T}}x : Ax \le b,\ x \ge 0\}$$
$$= \min\{b^{\mathsf{T}}y : A^{\mathsf{T}}y \ge c,\ y \ge 0\}.$$

So the meeting at 22 was no accident: *a perfect certificate always exists.* The bound-combining game can always be played to a draw.

STRONG DUALITY AS MINIMAX

Assume the standard-form primal has a finite optimum. With $x \geq 0$ and $y \geq 0$, $L(x, y) = c^{\mathsf{T}} x + y^{\mathsf{T}}(b - Ax)$.

Convention: write $\min = -\infty$ or $\max = +\infty$ when the expression is unbounded below or above, respectively.

- **Fix x, let y respond:** a violated row lets $y_i \to \infty$, so

$$\min_{y \geq 0} L(x, y) = \begin{cases} c^{\mathsf{T}} x & \text{if } Ax \leq b, \\ -\infty & \text{otherwise.} \end{cases}$$

  Hence $\max_x \min_y L$ is the **primal** optimum.

STRONG DUALITY AS MINIMAX

Now regroup by variable, as in Part II:

$$L(x, y) = b^{\mathsf{T}} y + (c - A^{\mathsf{T}} y)^{\mathsf{T}} x.$$

- **Fix y, let x respond:** a positive coefficient on x_j lets $x_j \to \infty$, so

$$\max_{x \geq 0} L(x, y) = \begin{cases} b^{\mathsf{T}} y & \text{if } A^{\mathsf{T}} y \geq c, \\ +\infty & \text{otherwise.} \end{cases}$$

  Hence $\min_y \max_x L$ is the **dual** optimum.

STRONG DUALITY AS MINIMAX

STRONG DUALITY, RESTATED If the primal has a finite optimum, then

$$\max_{x \geq 0} \min_{y \geq 0} L(x, y) = \min_{y \geq 0} \max_{x \geq 0} L(x, y).$$

- $\leq$ **is automatic** — that is weak duality: $\min_y L(x', y) \leq L(x', y') \leq \max_x L(x, y')$ for every x', y'.
- $=$ **is the deep part** (proof beyond this course). As a game: one player picks a production plan x, the other picks prices y. Strong duality says moving second gives no advantage.
- Von Neumann's minimax theorem for zero-sum games is a special case of LP duality.

THE FEASIBILITY CASES

Each LP has an optimum, is unbounded, or is infeasible. Of the nine pairings, only four occur:

PRIMAL \ DUAL	OPTIMUM	UNBOUNDED	INFEASIBLE
optimum	**possible** values equal	**never** weak duality	**never** strong duality
unbounded	**never** weak duality	**never** weak duality	**possible** forced by weak duality
infeasible	**never** strong duality	**possible** forced by weak duality	**possible** example next

A dual-feasible y caps every primal value at $b^{\mathsf{T}} y$ — so an unbounded side forces the other to be infeasible.

THE FEASIBILITY CASES

Both infeasible at once:

Primal

$$\begin{aligned} \max \quad & x_1 \\ & x_1 - x_2 \leq -1 \\ & -x_1 + x_2 \leq -1 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Dual

$$\begin{aligned} \min \quad & -y_1 - y_2 \\ & y_1 - y_2 \geq 1 \\ & -y_1 + y_2 \geq 0 \\ & y_1, y_2 \geq 0 \end{aligned}$$

- Primal: add the two rows to get $0 \leq -2$. Infeasible.
- Dual: add the two rows to get $0 \geq 1$. Infeasible.

PART IV — MAX-FLOW / MIN-CUT, ROUND TWO

RECALL WEEK 8

> **THEOREM (MAX-FLOW = MIN-CUT)** In any s–t network, the maximum flow value equals the minimum capacity of an s–t cut.

Week 8's proof: run Edmonds–Karp (Ford–Fulkerson with BFS) until no augmenting path exists; let S = vertices reachable from s in the *residual graph*; the cut $(S, V \setminus S)$ is saturated, so its capacity equals the flow value.

Today we re-prove the same theorem *without ever mentioning residual graphs* — it will fall out of LP duality.

MAX FLOW AS AN LP

Digraph $G = (V, E)$, capacities $c_e \geq 0$, source s, sink t (no edges into s).
One variable f_e per edge; $\delta^+(v)$ = edges out of v, $\delta^-(v)$ = edges into v:

$$\begin{aligned} \max \quad & \textstyle\sum_{e \in \delta^+(s)} f_e \\ \text{s.t.} \quad & f_e \leq c_e \quad \forall e \in E \\ & \textstyle\sum_{e \in \delta^-(v)} f_e \; - \; \sum_{e \in \delta^+(v)} f_e = 0 \quad \forall v \neq s, t \\ & f_e \geq 0 \quad \forall e \in E \end{aligned}$$

A maximisation with $\leq$ rows and $=$ rows. Penalise each row:

- capacity constraint of edge $e \rightarrow$ dual variable $y_e \geq 0$
- conservation constraint at vertex $v \rightarrow$ dual variable z_v, *free* (it's an equality)

DERIVING THE DUAL

Penalise every row, as in Part II:

$$L = \sum_{e \in \delta^+(s)} f_e + \sum_e y_e \left(c_e - f_e\right) + \sum_{v \neq s,t} z_v \left(\sum_{e \in \delta^+(v)} f_e - \sum_{e \in \delta^-(v)} f_e \right).$$

Then regroup by variable. Each f_e gets one coefficient; since $f_e \geq 0$, a finite bound needs that coefficient ≤ 0. **That is the dual constraint for edge e.**

ONE EDGE AT A TIME

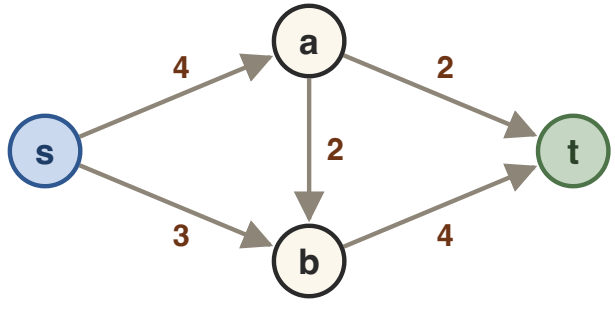
Take an edge $e = (u, v)$. Every term of L that contains f_e:

WHERE f_e APPEARS	ADDS TO ITS COEFFICIENT
objective, if $u = s$	$+1$
capacity row of e	$-y_e$
conservation at u **(**e **leaves** u**)**	$+z_u$
conservation at v **(**e **enters** v**)**	$-z_v$

Coefficient ≤ 0: $y_e \geq [u = s] + z_u - z_v$, with no z_s or z_t term (s and t have no conservation row).

ONE EDGE AT A TIME

Set $z_s := 1$ and $z_t := 0$. Then every edge, including those at s and t, reads the same way: $y_e \geq z_u - z_v$.



Week 9's network.

$$
\begin{aligned}
sa: &\quad y_{sa} \geq 1 - z_a \\
sb: &\quad y_{sb} \geq 1 - z_b \\
ab: &\quad y_{ab} \geq z_a - z_b \\
at: &\quad y_{at} \geq z_a \\
bt: &\quad y_{bt} \geq z_b
\end{aligned}
$$

Five edges, five dual constraints. What is left of L is $\sum_e c_e\, y_e$: minimise it.

DERIVING THE DUAL

DUAL LP

$$\begin{aligned} \min \quad & \textstyle\sum_e c_e\, y_e \\ \text{s.t.} \quad & y_e \;\geq\; z_u - z_v \quad \forall e = (u,v) \in E \\ & z_s = 1, \quad z_t = 0, \quad y_e \geq 0 \end{aligned}$$

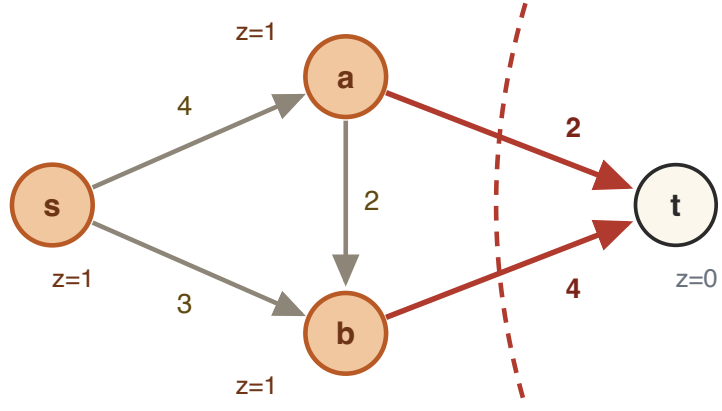
Intuition: pay c_e per unit of y_e; y_e must cover any *drop in the label* z along edge e; s is labelled 1, t is labelled 0.

READING THE DUAL: IT PRICES CUTS

Try a $0/1$ solution: pick $S \subseteq V$ with $s \in S, t \notin S$, and set $z_v = 1$ if $v \in S$, else 0.

Then $y_e \geq z_u - z_v$ forces $y_e \geq 1$ exactly on edges *leaving* S — and $y_e = 0$ works everywhere else.

Dual objective $= \sum_{e \text{ leaving } S} c_e$ = **the capacity of the cut S.**



Week 9's network. $S = \{s, a, b\}$: $z = 1$ on S, $z_t = 0$; $y_e = 1$ on the two red edges. Value $2 + 4 = 6$ — exactly last week's max flow. ($S = \{s\}$ gives $4 + 3 = 7$: a valid bound, but not tight.)

ROUNDING TO A CUT

Every cut is a $0/1$ dual solution, so the LP optimum is at most the min cut. For the other direction, take any optimal dual (y, z) and **round it to a cut** that costs no more.

- **Clip:** raise each negative z_v to 0, lower each $z_v > 1$ to 1.
  Still feasible: each new gap $z_u - z_v \le \max(0, \text{old gap}) \le y_e$.
- **Threshold:** pick θ uniformly from $(0, 1]$ and let

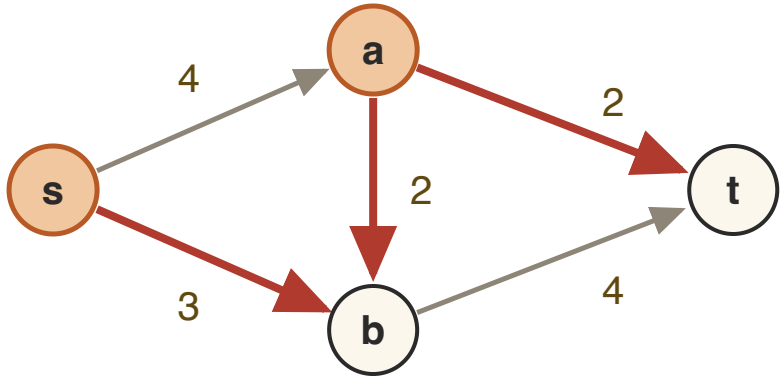
$$S_\theta = \{v : z_v \ge \theta\}.$$

  It contains s ($z_s = 1$) and never t ($z_t = 0$): an s–t cut.

THRESHOLD ROUNDING: A PICTURE

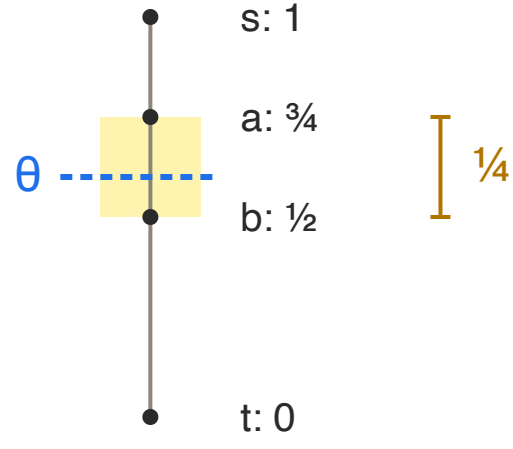
A feasible fractional dual: $z = (1, \frac{3}{4}, \frac{1}{2}, 0)$ on (s, a, b, t), with $y_{uv} = \max(0, z_u - z_v)$.

Capacities on edges



Orange: S_θ. Red: edges leaving S_θ.

Vertex potentials



Shaded interval: a → b crosses

Move θ: 0.60

S_θ = {s, a}, cut capacity = 7.

Edge $a \to b$ crosses exactly when $\frac{1}{2} < \theta \leq \frac{3}{4}$, so it crosses with probability $\frac{1}{4} = z_a - z_b$.

ROUNDING TO A CUT

For the proof, return to an optimal dual (y, z) after clipping.

- Edge $e = (u, v)$ leaves S_θ exactly when $z_v < \theta \le z_u$:

$$\Pr_\theta[e \text{ leaves } S_\theta] = \max(0,\ z_u - z_v) \ \le\ y_e.$$

- Sum over edges, weighted by capacity:

$$\begin{aligned}\mathbb{E}_\theta\big[\mathrm{cap}(S_\theta)\big] &= \textstyle\sum_e c_e \Pr_\theta[e \text{ leaves } S_\theta] \\ &\le \textstyle\sum_e c_e\, y_e \ =\ \text{LP optimum}.\end{aligned}$$

- Some θ does at least as well as the average: a cut of capacity $\le$ LP optimum.

ROUNDING TO A CUT

Cuts bound the LP from above, and rounding finds a cut no bigger than it. So the dual LP optimum *is* the min cut.

MAX-FLOW = MIN-CUT

$$\text{max flow} \overset{\text{strong duality}}{=} \text{dual LP optimum} \overset{\text{rounding}}{=} \text{min cut.}$$

SAME THEOREM, TWO PROOFS

	Week 8: residual graph	Today: LP duality
what you build	an algorithm (Ford–Fulkerson)	a certificate framework, no algorithm
integrality	integer capacities $\Rightarrow$ integer flow (integer augmentations)	$0/1$ cut labels via dual rounding, even for real capacities
scope	flows and close relatives	**every** linear program
the certificate	the cut found by the last BFS	any optimal dual solution

SAME THEOREM, TWO PROOFS

TAKEAWAY The residual-graph proof generalises to **algorithms** (Ford–Fulkerson); the LP proof generalises to **every LP**. Knowing both, you can pick the lens that fits the problem in front of you.

PART V — THE MATCHING CAPSTONE: KŐNIG'S THEOREM

THE MATCHING THREAD SO FAR

Week	What we did with matchings
W3	maximum matching on *trees* by DP
W4	greedy matching on trees + exchange arguments
W8	maximum *bipartite* matching via max flow
W9	the matching LP relaxation; integrality gaps
W10	**today: the capstone — duality explains it all**

The question left hanging since Week 8: when we found a maximum bipartite matching, *what certified that we could stop?*

DERIVING THE COVER DUAL

Graph $G = (V, E)$; variable x_e per edge (how much of e we take). The matching LP:

$$\max \ \sum_{e \in E} x_e \quad \text{s.t.} \quad \sum_{e \ni v} x_e \le 1 \ \ \forall v \in V, \quad x \ge 0.$$

Penalise each vertex row with a price y_v. It is a $\le$ row, so $y_v \ge 0$:

$$L = \sum_e x_e \ + \ \sum_v y_v \Big(1 - \sum_{e \ni v} x_e\Big).$$

Then regroup by variable: the coefficient of each x_e must be ≤ 0.

ONE EDGE AT A TIME

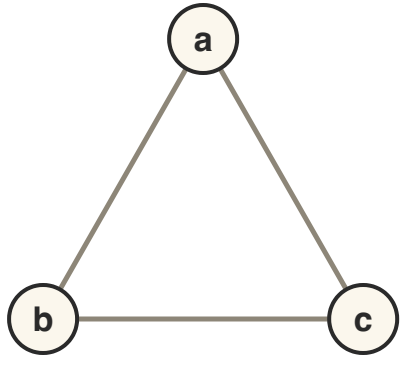
Take an edge $e = uv$. Every term of L that contains x_e:

WHERE x_e APPEARS	ADDS TO ITS COEFFICIENT
objective	$+1$
vertex row of u	$-y_u$
vertex row of v	$-y_v$
any other vertex row	nothing: e doesn't touch it

Coefficient $1 - y_u - y_v \leq 0$: $\quad y_u + y_v \geq 1$.
What is left of L is $\sum_v y_v$: minimise it.

ONE EDGE AT A TIME

Week 9's triangle: three vertex rows give $y_a, y_b, y_c \geq 0$; three edges give three dual constraints.



$$\begin{aligned} \min \quad & y_a + y_b + y_c \\ ab: \quad & y_a + y_b \geq 1 \\ ac: \quad & y_a + y_c \geq 1 \\ bc: \quad & y_b + y_c \geq 1 \end{aligned}$$

Each edge must be covered by its two endpoints: a **fractional vertex cover**.

MATCHING $\leftrightarrow$ VERTEX COVER

Matching LP (primal)

$$\begin{aligned} \max \quad & \textstyle\sum_{e \in E} x_e \\ & \textstyle\sum_{e \ni v} x_e \le 1 \quad \forall v \in V \\ & x_e \ge 0 \end{aligned}$$

Its dual (derived)

$$\begin{aligned} \min \quad & \textstyle\sum_{v \in V} y_v \\ & y_u + y_v \ge 1 \quad \forall uv \in E \\ & y_v \ge 0 \end{aligned}$$

The dual is the **fractional vertex cover** LP: pay for vertices so that every edge is covered at least once. The two problems we kept meeting are *LP duals*.

WEAK DUALITY, COMBINATORIALLY

OBSERVATION For *any* matching M and *any* vertex cover C in *any* graph: $|M| \leq |C|$.

Why: each edge of M must contain at least one vertex of C; and no vertex of C can account for two edges of M, since matching edges are disjoint. So C has at least one private vertex per matching edge. ■

This is exactly the weak duality inequality $c^{\mathsf{T}}x \leq b^{\mathsf{T}}y$, specialised to $0/1$ solutions of the two LPs.

WEAK DUALITY, COMBINATORIALLY

So: a matching and a cover *of the same size* would certify each other. Do they always exist? In general graphs, no (triangle: max matching 1, min cover 2). But…

BIPARTITE: THE GAP CLOSES

Fact (integrality — Week 9) If G is **bipartite**, the matching LP and the vertex cover LP both have optimal solutions that are $0/1$-valued.

(Week 9: the incidence matrix of a bipartite graph is *totally unimodular*, so by Cramer's rule every polytope vertex is integral.)

BIPARTITE: THE GAP CLOSES

Recall Week 9's warning example: on the triangle C_3, setting $x_e = \frac{1}{2}$ on all three edges gives matching-LP value $\frac{3}{2} > 1$ — an integrality gap. Odd cycles are the culprit, and bipartite graphs have none.

Now chain the equalities:

$$\begin{aligned} \text{max matching} &\overset{\text{integrality}}{=} \text{LP}_{\text{match}} \\ &\overset{\text{strong duality}}{=} \text{LP}_{\text{cover}} \\ &\overset{\text{integrality}}{=} \text{min vertex cover}. \end{aligned}$$

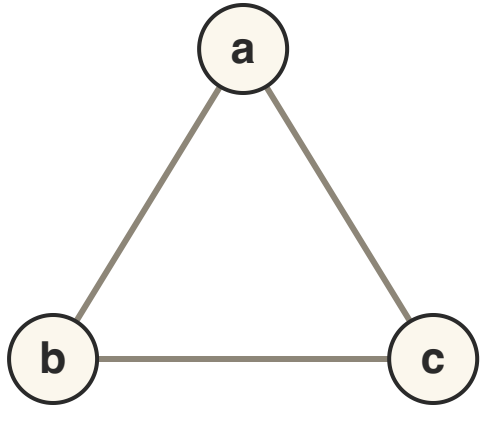
KŐNIG'S THEOREM

Theorem (Kőnig, 1931) In every bipartite graph, the size of a maximum matching equals the size of a minimum vertex cover.

- The chain on the previous slide *is* the proof — duality plus integrality, no new combinatorics needed.
- Practical upshot: a matching M and a cover C with $|M| = |C|$ form a **mutual certificate** — hand both to the skeptic.

THE TRIANGLE

Which equalities from the bipartite chain survive on K_3?



Compute all four optima:

1. Maximum matching
2. Maximum fractional matching
3. Minimum fractional cover
4. Minimum vertex cover

THE TRIANGLE: DUALITY SURVIVES

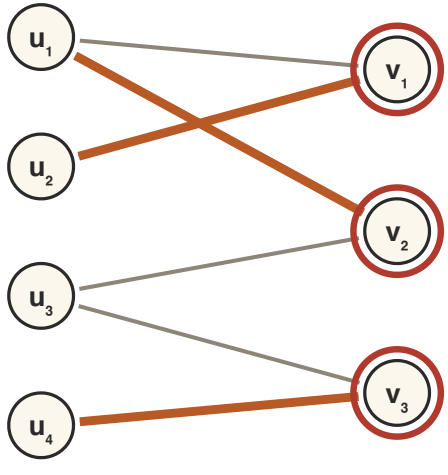
$$\underbrace{1}_{\substack{\max\\ \text{matching}}} < \underbrace{\tfrac{3}{2}}_{\text{matching LP}} = \underbrace{\tfrac{3}{2}}_{\text{cover LP}} < \underbrace{2}_{\substack{\min\\ \text{cover}}}.$$

Put $x_e = \frac{1}{2}$ on every edge and $y_v = \frac{1}{2}$ on every vertex. Both LP solutions are feasible and have value $\frac{3}{2}$, so weak duality certifies that both are optimal.

Any two edges share a vertex, so a matching has size at most 1. A vertex cover needs 2 vertices.

> **WHAT BREAKS?** Strong duality still gives the middle equality. Both **integrality** equalities fail.

WORKED EXAMPLE: A MATCHING CERTIFIES A COVER



Orange edges: matching M. Red rings: cover $C = \{v_1, v_2, v_3\}$.

Bipartite graph: edges $u_1v_1, u_1v_2, u_2v_1, u_3v_2, u_3v_3, u_4v_3$.

Matching
$M = \{u_1v_2,\ u_2v_1,\ u_4v_3\}$, size $\mathbf{3}$ (only u_3 unmatched).

Cover $C = \{v_1, v_2, v_3\}$, size $\mathbf{3}$: every edge ends on the right, so all six are covered.

WORKED EXAMPLE: A MATCHING CERTIFIES A COVER

MUTUAL CERTIFICATION Any cover needs $\geq |M| = 3$ vertices, so C is a minimum cover; any matching has $\leq |C| = 3$ edges, so M is a maximum matching. Each is the other's proof.

THE VERTEX COVER ARC

Graph	Min vertex cover	When
trees	easy — DP / Gallai	Week 3
bipartite	polynomial — Kőnig: reduce to matching	Week 10 (today)
general graphs	…what do you think?	Week 11

Duality gave us the bipartite case as a two-line corollary. For general graphs the matching–cover gap reopens (remember the triangle) — and something much deeper goes wrong. That story is **NP-completeness, next week**.

PART VI — COMPLEMENTARY SLACKNESS

COMPLEMENTARY SLACKNESS

When do a primal-feasible x and dual-feasible y actually meet, $c^\mathsf{T} x = b^\mathsf{T} y$?

THEOREM (COMPLEMENTARY SLACKNESS) Feasible x and y are *both optimal* if and only if:

- for every j: $x_j > 0 \;\Rightarrow\; (A^\mathsf{T} y)_j = c_j$ (used variable $\Rightarrow$ its dual constraint is tight)
- for every i: $y_i > 0 \;\Rightarrow\; (Ax)_i = b_i$ (paid-for constraint $\Rightarrow$ it is tight)

COMPLEMENTARY SLACKNESS

Slogan: at the optimum, nothing is both *used* and *slack*. You never pay a positive price y_i for a constraint you aren't pressing against, and you never run a positive activity x_j whose dual price exceeds its value.

WHY IT HOLDS: SQUEEZE THE WEAK DUALITY CHAIN

Recall the one-line proof:

$$c^{\mathsf{T}}x \overset{(1)}{\leq} y^{\mathsf{T}}Ax \overset{(2)}{\leq} y^{\mathsf{T}}b.$$

If x and y are both optimal, strong duality says the two ends are *equal* — so (1) and (2) are equalities.

WHY IT HOLDS: SQUEEZE THE WEAK DUALITY CHAIN

Both $\leq$ in $c^\mathsf{T}x \overset{(1)}{\leq} y^\mathsf{T}Ax \overset{(2)}{\leq} y^\mathsf{T}b$ are equalities at an optimal pair. Now read each termwise.

Inequality (1) written termwise: $\sum_j x_j\big((A^\mathsf{T}y)_j - c_j\big) = 0$, a sum of *nonnegative* terms. A nonnegative sum is zero only if every term is zero: $x_j > 0$ forces $(A^\mathsf{T}y)_j = c_j$.

Same for (2): $\sum_i y_i\big(b_i - (Ax)_i\big) = 0$ gives $y_i > 0 \Rightarrow (Ax)_i = b_i$. And conversely, if all terms vanish the chain collapses to equality, so both are optimal. ■

WORKED CHECK: THE RUNNING EXAMPLE

Claimed optimal pair: $x^\star = (6, 2)$ and $y^\star = (1, 2, 0)$. First, the two primal variables:

CONDITION	CHECK	
$x_1 = 6 > 0$	dual constraint 1 tight? $y_1 + y_2 + y_3 = 1 + 2 + 0 = 3 = c_1$	✓
$x_2 = 2 > 0$	dual constraint 2 tight? $y_2 + 3y_3 = 2 + 0 = 2 = c_2$	✓

WORKED CHECK: THE RUNNING EXAMPLE

Then the three dual variables, one per resource:

CONDITION	CHECK	
$y_1 = 1 > 0$	machine tight? $x_1 = 6 = b_1$	✓
$y_2 = 2 > 0$	wood tight? $x_1 + x_2 = 6 + 2 = 8 = b_2$	✓
$y_3 = 0$	varnish is slack, $x_1 + 3x_2 = 12 < 18$ — allowed only because $y_3 = 0$	✓

WORKED CHECK: THE RUNNING EXAMPLE

All conditions hold, and both points are feasible $\Rightarrow$ both optimal — *without computing either objective value*.

The varnish row is the one with content: a constraint with slack *must* get multiplier 0. Attempt 1 put weight on varnish, and that is exactly why it could not reach 22.

SOLVING FOR THE CERTIFICATE

Given only the claimed optimum $x^\star = (6, 2)$, complementary slackness hands us equations for $y^\star$:

- varnish is slack $\Rightarrow y_3 = 0$;
- $x_2 > 0 \Rightarrow y_2 + 3y_3 = 2$, so $y_2 = 2$;
- $x_1 > 0 \Rightarrow y_1 + y_2 + y_3 = 3$, so $y_1 = 1$.

Check $y^\star = (1, 2, 0) \geq 0$: it is dual feasible, so $(6, 2)$ is certified. (These equations pin $y^\star$ down uniquely, so a negative entry would have meant $(6, 2)$ is not optimal.)

WHEN A PRICE GOES NEGATIVE

Try complementary slackness at a vertex that is *not* optimal: $x = (3, 5)$, value 19. Wood and varnish are tight; the machine is slack.

- machine slack $\Rightarrow y_1 = 0$;
- $x_1, x_2 > 0 \Rightarrow y_2 + y_3 = 3$ and $y_2 + 3y_3 = 2$;
- so $y_2 = \frac{7}{2}$ and $y_3 = -\frac{1}{2}$.

The combination $\frac{7}{2}\,(\text{wood}) - \frac{1}{2}\,(\text{varnish})$ gives exactly 19 — but subtracting a $\leq$ constraint is not allowed. **No certificate exists at $(3, 5)$.**

WHEN A PRICE GOES NEGATIVE

The negative price on varnish says: *loosening varnish pays*.

- Keep wood tight and move off varnish: trade one table for one chair, $(3, 5) \to (4, 4)$. Varnish loosens by 2, and the objective gains $(-\frac{1}{2})(-2) = +1$.
- Continue until the machine becomes tight: $(6, 2)$, value 22.
- There the prices are $(1, 2, 0) \geq 0$: a valid certificate, so stop.

This illustrates how the **simplex method** uses prices to choose pivots. To certify a feasible point, check **dual feasibility** ($y \geq 0$ and $A^{\mathsf{T}} y \geq c$) together with complementary slackness. (Week 12's *primal–dual* algorithms use complementary slackness as a design principle too.)

TAKEAWAYS

- Every LP has a **dual**: the LP of best bounds obtained by combining constraints. To write it for any LP, penalise each constraint with a price y_i: the signs and dual constraints are whatever keeps the bound valid and finite.
- **Weak duality** ($c^\mathsf{T} x \leq b^\mathsf{T} y$) is a one-line proof; **strong duality** says every finite primal optimum has a matching dual certificate.
- **Max-flow = min-cut** is LP duality: the dual of the flow LP prices cuts. Two proofs, two generalisations — algorithms vs. all of LP.

TAKEAWAYS

- **Kőnig:** in bipartite graphs, max matching $=$ min vertex cover — duality $+$ integrality; matching and cover certify each other.
- **Complementary slackness:** nothing optimal is both used and slack. Check primal and dual feasibility too; in our example, a negative price points to an improving move.